

Design Optimization of a Near-Shore Smart Buoy Hull

Group 14 Final Project

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March 22, 2026

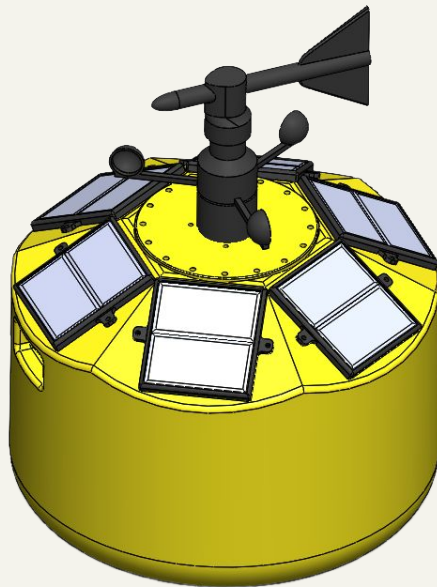
Near-Shore Smart Buoy

Ocean monitoring buoys are used for:

- Wave measurement
- Environmental monitoring
- Navigation and safety

Current buoy design lacks:

- Validated structural analysis
- Stability verification
- Structural optimization



Structures must withstand:

- Wave loading
- Mooring tension
- Hydrodynamic forces

Key engineering questions

- Is the hull structurally safe?
- How large are mooring-induced loads?
- Can the structure be lightened or reduced in cost without failure risk?

Project Objectives



Stability/Loading Analysis

- Verification of Floating Stability
- Determination of Mooring and Hydrostatic Forces

Structural Analysis

- Validate Current Hull Design Subject to External Forces
- Validate Multi Piece 3D Printed Hull

Structural Optimization

- Perform Sensitivity Analysis
- Minimize Hull thickness Subject to Stress Constraint

NX Model



GRANTA EDUPACK

PLA (general purpose)

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General information

Designation

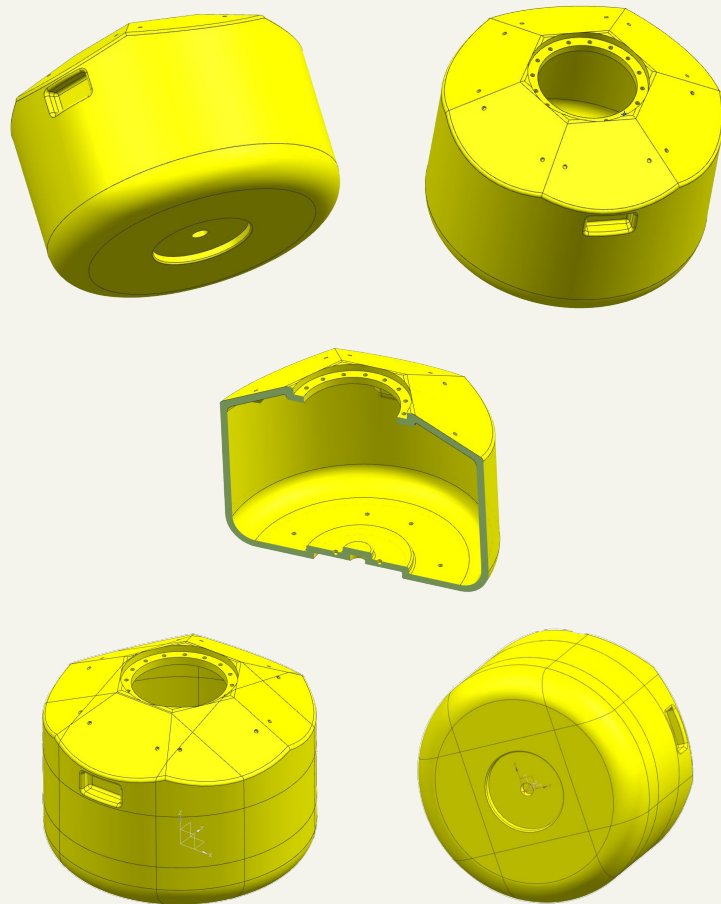
PLA (general purpose), Polylactide / Polylactic acid (General purpose)

Physical properties

Density	1.2e3	-	1.29e3	kg/m ³
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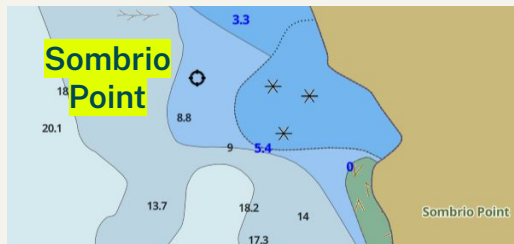
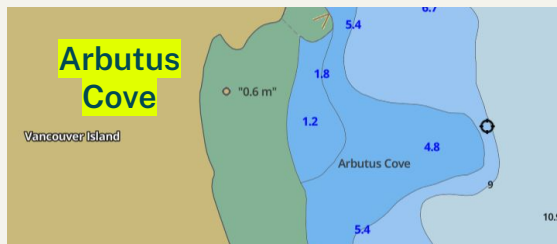
Mechanical properties

Young's modulus	2.4	-	3.6	GPa
Specific stiffness	1.92	-	2.9	MN.m/kg
Yield strength (elastic limit)	45	-	72	MPa
Tensile strength	45	-	72	MPa
Specific strength	36.1	-	58	kN.m/kg



Hydrodynamic Simulation

Deployment Locations



During large pacific winter storms:

Douglas Sea Scale Degree	Height (m)	Description
0	No wave	Calm (Glassy)
1	0 – 0.1	Calm (Rippled)
2	0.1 – 0.5	Smooth
3	0.5 – 1.25	Slight
4	1.25 – 2.5	Moderate
5	2.5 – 4	Rough
6	4 – 6	Very Rough
7	6 – 9	High
8	9 - 14	Very High
9	14 +	Phenomenal

Near shore conditions (approximately 100-300m):

0.5
m/s

Current
speed

6
m

Wave
Height

15
s

Wave
Period

40
km/h

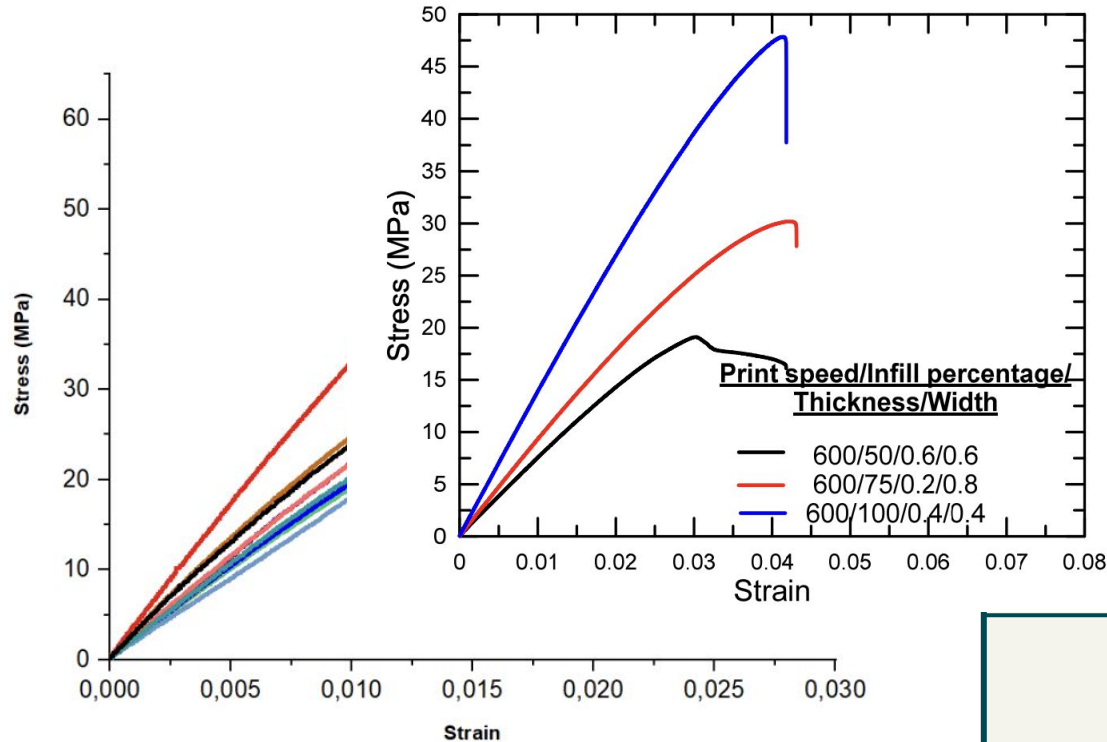
Wind
Speed

Hydrodynamic Simulation



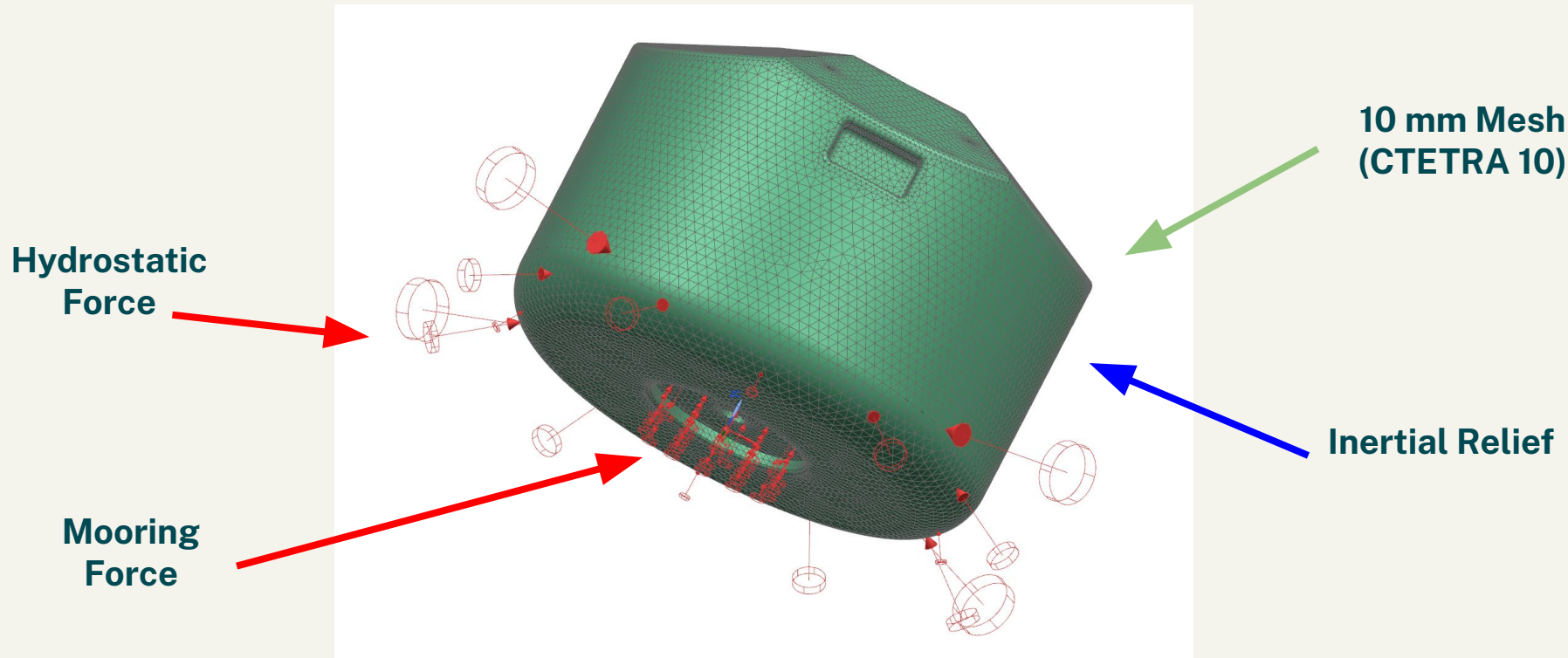
Maximum tension
 $\approx 340 \text{ N}$

PLA Infill Assumption and Effect on Strength



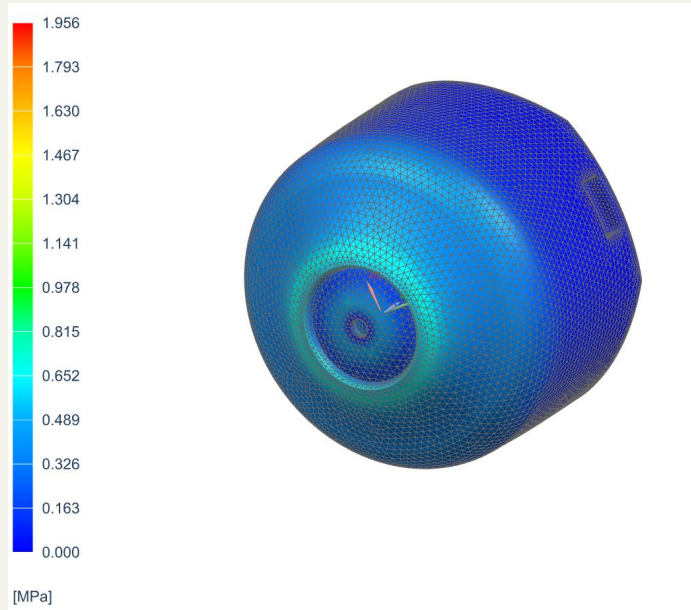
<u>Infil</u>	<u>Stress (MPa)</u>
100 %	45
75 %	30
<u>Infill Pattern</u>	<u>Stress (MPa)</u>
Hexagon	3
⋮	
25 %	6

NX Loads & Constraints

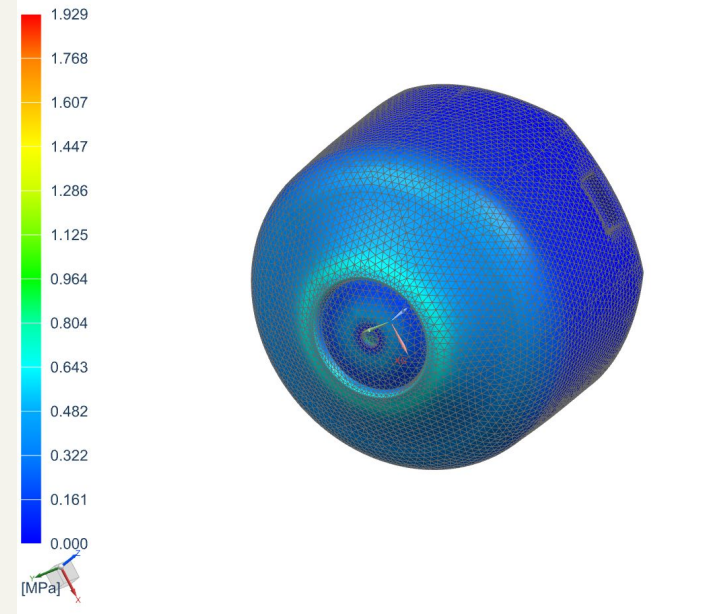


Structural Analysis

Injection Molded PE

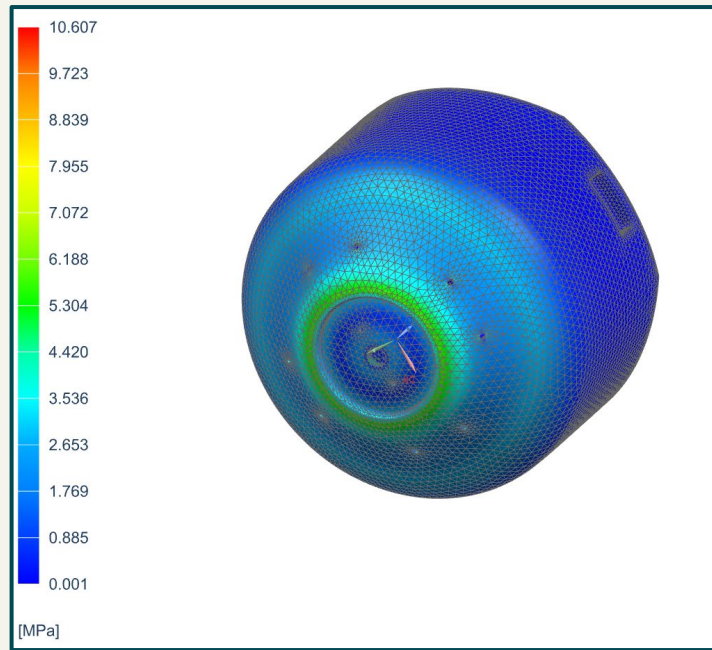


PLA



Sensitivity & Optimization

Revised Max Stress



Wall Thickness Change

12.7mm

4.58mm

Stress Change

1.95MPa

10.6MPa

Material Savings

62.4%*

**By volume over PLA*

Thank You!

Q&A
